Uncertainty relations from majorization and entropy in the binary case

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(Dated: October 25, 2026)

Making one quantum measurement more predictable can force another to become less so. Majorization expresses this trade-off with a single probability vector, valid for every state, that bounds a whole family of uncertainty measures at once. We study the simplest case: two measurements with two outcomes each. The optimal such vectors are known; we prove that no state attains them unless the measurements commute. Every bound they imply for a strictly Schur-concave measure, such as the Shannon entropy, is therefore strict; only the Rényi entropies of order zero and infinity are bounded tightly, and the direct-sum form fails at large orders. The Shannon bound still beats the Maassen–Uffink relation unless the measurements are nearly mutually unbiased, and becomes tight in ratio, though slowly, as they approach commuting ones. We also derive the optimal bound for noisy and biased measurements and extend the results to any finite dimension.

I. INTRODUCTION

The uncertainty principle is one of the defining features of quantum mechanics [1–3]: there are pairs of measurements whose outcomes cannot both be predicted with certainty, however carefully the system is prepared. This preparation uncertainty is distinct from measurement uncertainty, which limits how accurately incompatible observables can be measured jointly [4]; we are concerned with the former. First formulated for the position and momentum of a particle, the principle has since become a quantitative tool of quantum information science. It underlies the security of quantum key distribution [5–7], provides experimentally accessible criteria for entanglement [6, 8, 9], and is closely related to, though distinct from, the incompatibility of quantum measurements [10, 11]. These applications call for formulations that apply to measurements with a finite number of outcomes and whose bounds do not depend on the state of the system.

Robertson's variance-based relation [2] does not meet these requirements: its lower bound depends on the state, it can vanish even for incompatible observables, and variances are not natural when the outcomes are mere labels [12]. Entropies of the outcome distributions provide a remedy [12–16]. The best known entropic uncertainty relation, due to Maassen and Uffink [17], bounds the sum of the Shannon entropies of the outcomes of two measurements by a quantity that depends only on the largest overlap between their eigenvectors. Entropic uncertainty relations have since grown into a broad field with applications throughout quantum information theory [18, 19]; they have been refined in various regimes [20, 21] and tested in experiments [22, 23].

Each such relation, however, constrains a single measure of uncertainty, while different tasks call for different measures: the Shannon entropy quantifies average information, the min-entropy governs guessing probabilities [24], the Rényi family interpolates between these and other extremes, and the Tsallis family generalizes the Shannon entropy in a different way. Majorization [25], the order that captures when one probability distribution is more disordered than another, makes it possible to treat all of them at once. It is a central tool of quantum information theory, where it governs, for instance, the transformations of entangled states [26] and, more generally, quantum resource theories [27, 28]. In the context of uncertainty, the idea is to find a single probability vector, independent of the state, that majorizes the outcome statistics of every state; every uncertainty measure that respects majorization then obeys a bound determined by this one vector. Depending on whether the two outcome distributions are combined by a direct sum or by a tensor product, such measures include the Shannon and Tsallis entropies or the Shannon and all Rényi entropies, among many others. This program was initiated by Partovi [29] and developed into universal uncertainty relations by Friedland, Gheorghiu and Gour [30] and by Puchała, Rudnicki and Życzkowski [31]; Rudnicki *et al.* subsequently introduced a direct-sum formulation that yields stronger entropic bounds [32]. Majorization uncertainty relations have since been extended to generalized measurements and quantum operations [33, 34], sharpened further [35–38], in particular by exploiting the lattice structure of majorization, and verified experimentally [38, 39].

The price of this universality is only partly understood. For a qubit measured in two bases—the setting of many experiments and, for mutually unbiased bases, of the BB84 protocol [5]—entropy-specific relations can be optimized: the optimal Shannon relation is known [40, 41], in closed form over part of the range and through a one-

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parameter minimization otherwise, and so are optimal relations for Rényi entropies [42, 43]. Earlier majorization bounds have been compared with the Maassen–Uffink relation and with the optimal Shannon bound [34, 44], and the optimal direct-sum bound for two qubit bases has been identified in the majorization lattice [36]. Li and Qiao also observed that the partial sums of this bound are usually reached by different states and that, in examples, its Shannon entropy falls short of the optimal Shannon bound [45]. What is missing is a complete account of when such bounds are tight, why, and how this depends on the uncertainty measure. In the simplest setting, where both approaches can be carried out completely, what exactly does majorization gain, and what does it cost?

In this paper we answer this question for pairs of binary (two-outcome) measurements, and the answer is simple to state. Unless the measurements commute, no state attains the optimal majorization bound, so every bound it implies for a strictly Schur-concave measure of uncertainty, such as the Shannon entropy, is strict; only the Rényi entropies of order zero and infinity are bounded tightly. The price is substantial: for two qubit bases the direct-sum Shannon bound falls short of the exact minimum by up to 22%. Even so, it outperforms the Maassen–Uffink relation unless the measurement axes on the Bloch sphere are within about nine degrees of orthogonal, i.e., unless the measurements are nearly mutually unbiased [34], and its ratio to the exact minimum tends to one, though only logarithmically, as the measurements approach compatibility, where the Maassen–Uffink relation is weakest. For the Shannon entropy alone, a known closed-form bound [40, 44] is stronger than both at every angle; the case for majorization rests on its universality, not on its sharpness for any single measure.

Several ingredients of this analysis are known. For two qubit bases the relevant majorization bounds are special cases of the general constructions of Refs. [30–32]; the optimality of the direct-sum bound and the states that saturate its partial sums were found in Ref. [36]; the expectation values that qubit states can reach are known to fill an ellipse [44, 46, 47]; and for unbiased unsharp qubit observables the direct-sum bound was evaluated in Ref. [34]. We give a unified and self-contained treatment that derives both bounds, their optimality and the states that saturate each of their partial sums from this ellipse. What is new is a proof of the non-attainment result for both bounds, which makes rigorous and general the observations of Ref. [45] on the direct-sum bound, and its consequence, the strictness of all induced bounds for strictly Schur-concave measures, with only Rényi orders zero and infinity tight; a quantitative account of the resulting gap; the comparison of both types of majorization bound with the Maassen–Uffink relation and with the exact optimum across the whole Rényi family, including an explicit counterexample to the direct-sum form at large orders, whose failure for $\alpha > 1$ was stated without proof in Ref. [32]; the optimal direct-sum bound for arbitrary noisy and biased two-outcome qubit measurements, the conditions under which it is attained, and its relation to joint measurability, including a trade-off of classical origin that bias creates between commuting measurements; and the extension to binary measurements in any finite dimension, with both bounds for projective measurements and the direct-sum bound for noisy and biased ones.

The paper is organized as follows. Section II introduces majorization, the uncertainty measures and the measurements we consider, and Sec. III describes the geometry of the achievable statistics. Section IV derives the optimal bounds for projective qubit measurements, and Sec. V discusses their entropic consequences and compares them with entropy-specific relations. Section VI treats general two-outcome qubit measurements, which may be noisy or biased and are described by positive operator-valued measures (POVMs), i.e., sets of positive operators that sum to the identity; Sec. VII treats binary measurements in arbitrary dimension. Section VIII discusses the results, including their relation to quantum optics, and concludes. All analytical results were also checked numerically (Appendix A).

II. PRELIMINARIES

A. Majorization and uncertainty measures

For $x \in \mathbb{R}^n$ let $x^{\downarrow}$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^{k} x_i^{\downarrow}$ its Lorenz partial sums, with $L_0(x) = 0$; the Lorenz curve of x is the piecewise-linear interpolation of the points $(k, L_k(x))$, $k = 0, \ldots, n$. For x, y with equal total, x is *majorized* by y, written $x \prec y$, if $L_k(x) \le L_k(y)$ for all k [25]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \ge \Phi(y)$, and *strictly* Schur-concave if, in addition, $\Phi(x) > \Phi(y)$ whenever $x \prec y$ and $x^{\downarrow} \neq y^{\downarrow}$. Every function $\Phi(x) = \sum_i \phi(x_i)$ with ϕ (strictly) concave is (strictly) Schur-concave [25].

The uncertainty measures used below are the Rényi [48] and Tsallis [49] entropies

$$H_\alpha(p) = \frac{\log_2 \sum_i p_i^\alpha}{1-\alpha}, \qquad T_\gamma(p) = \frac{1 - \sum_i p_i^\gamma}{\gamma - 1}, \tag{1}$$

with $\alpha \in [0, \infty]$ and $\gamma > 0$; the cases $\alpha \in \{0, 1, \infty\}$ and $\gamma = 1$ are defined by continuity, with the Shannon entropy $H = H_1$ as the common limit ($T_\gamma \to H \ln 2$ as $\gamma \to 1$). The orders $\alpha = \infty$ and $\alpha = \frac{1}{2}$ give the min-entropy $H_\infty(p) = -\log_2 \max_i p_i$ and the max-entropy $H_{1/2}$ [24], and H_0 is the logarithm of the number of non-zero entries. All of them are Schur-concave; H_α is strictly Schur-concave for $0 < \alpha < \infty$ and T_γ for all $\gamma > 0$, whereas H_0 and H_∞ are not.

Let $p(\rho)$ and $q(\rho)$ denote the outcome distributions of two measurements M and N on a state ρ; we also write $H_\alpha(M) = H_\alpha(p)$ and $H_\alpha(N) = H_\alpha(q)$. For measurements in orthonormal bases $\{|m_i\rangle\}$ and $\{|n_j\rangle\}$, the

Maassen–Uffink (MU) relation [17] reads

$$H(p) + H(q) \;\geq\; -\log_2 c, \qquad c = \max_{i,j} |\langle m_i | n_j \rangle|^2. \quad (2)$$

It constrains one particular functional of the pair (p, q); a different uncertainty measure requires a different relation. A majorization uncertainty relation [29–32] instead consists of a single state-independent vector ω such that

$$p(\rho) \oplus q(\rho) \;\prec\; \omega \quad \text{or} \quad p(\rho) \otimes q(\rho) \;\prec\; \omega \qquad \forall \rho. \quad (3)$$

We call these the direct-sum and the tensor-product form, respectively. Every Schur-concave function Φ then inherits the bound $\Phi(p \oplus q) \geq \Phi(\omega)$ or $\Phi(p \otimes q) \geq \Phi(\omega)$. Sums of Shannon or Tsallis entropies are Schur-concave functions of $p \oplus q$, since, the totals of p and q being fixed, they are sums of a concave function over all entries; sums of Shannon or Rényi entropies are Schur-concave functions of $p \otimes q$, since $H_\alpha(p \otimes q) = H_\alpha(p) + H_\alpha(q)$. The direct-sum form also yields bounds for Rényi entropies of order $\alpha \leq 1$, by a separate argument (Corollary 1), but not for all orders (Proposition 2).

Restricted to non-negative, non-increasing vectors of fixed length and fixed total, $\prec$ is a partial order, and in fact a complete lattice [50, 51].

Definition 1. Let $\mathcal{A}$ be a set of non-negative vectors of common length and total. A vector ω is an *optimal majorization bound* for $\mathcal{A}$ if $a \prec \omega$ for all $a \in \mathcal{A}$, and $\omega \prec \omega'$ for every ω' with this property; i.e., $\omega = \sup \mathcal{A}$ in the majorization lattice.

Lemma 1 ([50, 51])**.** *Let $S_k = \sup_{a \in \mathcal{A}} L_k(a)$. Then $\sup \mathcal{A}$ is the vector whose Lorenz curve is the least concave majorant of $k \mapsto S_k$ (with $S_0 = 0$). In particular, if $\omega_k = S_k - S_{k-1}$ is already non-increasing, then $\omega = \sup \mathcal{A}$.*

In all cases treated below the second, simpler statement applies. The quantities S_k have a transparent meaning: S_k is the largest value that any state can give to a sum of k entries, i.e., to the summed probabilities of k outcomes.

B. Two-outcome qubit measurements

A qubit state is $\rho = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma})$ with $|\vec{r}| \leq 1$. A projective two-outcome measurement is $M = \{P_\pm\}$ with $P_\pm = \frac{1}{2}(\mathbb{1} \pm \hat{m} \cdot \vec{\sigma}) = |m_\pm\rangle\langle m_\pm|$, $|\hat{m}| = 1$, and similarly $N = \{Q_\pm\}$ with axis $\hat{n}$ and eigenvectors $|n_\pm\rangle$. The outcome distributions are

$$p = \left(\tfrac{1+x}{2}, \tfrac{1-x}{2}\right), \qquad q = \left(\tfrac{1+y}{2}, \tfrac{1-y}{2}\right), \quad (4)$$

where $x = \hat{m} \cdot \vec{r}$ and $y = \hat{n} \cdot \vec{r}$. The overlaps are $|\langle m_\pm | n_\pm \rangle|^2 = \frac{1}{2}(1 + \hat{m} \cdot \hat{n})$ and $|\langle m_\pm | n_\mp \rangle|^2 = \frac{1}{2}(1 - \hat{m} \cdot \hat{n})$, hence

$$c = \tfrac{1}{2}\big(1 + |\hat{m} \cdot \hat{n}|\big) = \cos^2 \tfrac{\theta}{2}, \qquad \cos\theta = |\hat{m} \cdot \hat{n}|, \quad (5)$$

where $\theta \in [0, \frac{\pi}{2}]$ is the (acute) Bloch angle between the two measurement axes. Relabeling the outcomes of N maps $\hat{n} \mapsto -\hat{n}$ and does not affect any majorization statement, so we may and shall assume $\hat{m} \cdot \hat{n} = \cos\theta \geq 0$. Note that $\frac{1}{2} \leq c \leq 1$ and $\sqrt{c} \geq 1/\sqrt{2} > 1 - \sqrt{c}$.

A general two-outcome POVM $\{E_0, E_1 = \mathbb{1} - E_0\}$ can always be written as

$$E_0 = \tfrac{1}{2}\big[(1 + \mu)\mathbb{1} + \eta\, \hat{a} \cdot \vec{\sigma}\big], \qquad \eta \geq 0,\ |\mu| + \eta \leq 1, \quad (6)$$

with $|\hat{a}| = 1$, the constraint being equivalent to $0 \leq E_0 \leq \mathbb{1}$. Here η is the sharpness and μ the bias; rank-one projective measurements correspond to $\eta = 1$, $\mu = 0$.

III. GEOMETRY OF THE ACHIEVABLE SET

The statistics (4) depend on $\vec{r}$ only through (x, y).

Lemma 2 ([44, 46, 47])**.** *For non-parallel axes, the set $\{(\hat{m} \cdot \vec{r}, \hat{n} \cdot \vec{r}) : |\vec{r}| \leq 1\}$ is the filled ellipse*

$$\mathcal{E}_\theta = \big\{(x, y) : x^2 - 2xy\cos\theta + y^2 \leq \sin^2\theta\big\}. \quad (7)$$

Its boundary consists exactly of the images of the pure states whose Bloch vectors lie in $\mathrm{span}\{\hat{m}, \hat{n}\}$*. Moreover, for all $s, t \in \mathbb{R}$,*

$$\max_{|\vec{r}| \leq 1} (sx + ty) = |s\hat{m} + t\hat{n}|. \quad (8)$$

Proof. Write $\vec{r} = \vec{r}_\parallel + \vec{r}_\perp$ with $\vec{r}_\parallel \in \mathrm{span}\{\hat{m}, \hat{n}\}$. Only $\vec{r}_\parallel$ contributes to (x, y), and $(x, y) = G\,\xi$ where ξ are the coefficients of $\vec{r}_\parallel$ in the basis $\{\hat{m}, \hat{n}\}$ and $G = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$ is the Gram matrix. Then $|\vec{r}_\parallel|^2 = \xi^T G \xi = (x, y) G^{-1} (x, y)^T \leq 1$, which is the stated inequality since $\det G = \sin^2\theta$; equality holds iff $\vec{r}_\perp = 0$ and $|\vec{r}| = 1$. Conversely, every $(x, y) \in \mathcal{E}_\theta$ is attained by $\vec{r} = \xi_1 \hat{m} + \xi_2 \hat{n}$ with $\xi = G^{-1}(x, y)^T$, since then $|\vec{r}|^2 = (x, y) G^{-1} (x, y)^T \leq 1$. Eq. (8) is Cauchy–Schwarz: $sx + ty = \vec{r} \cdot (s\hat{m} + t\hat{n})$. □

Equivalently, $(x, y) \in \mathcal{E}_\theta$ if and only if the covariance matrix of the observables $A = \hat{m} \cdot \vec{\sigma}$ and $B = \hat{n} \cdot \vec{\sigma}$,

$$\Gamma = \begin{pmatrix} 1 - x^2 & \cos\theta - xy \\ \cos\theta - xy & 1 - y^2 \end{pmatrix}, \quad (9)$$

is positive semidefinite, because $\det\Gamma = \sin^2\theta - (x^2 - 2xy\cos\theta + y^2)$. Its off-diagonal entry contains the anticommutator $\frac{1}{2}\{A, B\} = \cos\theta\,\mathbb{1}$, which does not depend on the state. The ellipse is a disk for mutually unbiased measurements ($\theta = \pi/2$) and degenerates into the segment $x = y$ as $\theta \to 0$ [47].

Consequently the achievable set $\mathcal{P}_\oplus(M, N) = \{p(\rho) \oplus q(\rho)\} \subset \mathbb{R}^4$ is the image of $\mathcal{E}_\theta$ under the affine map $(x, y) \mapsto \frac{1}{2}(1 + x, 1 - x, 1 + y, 1 - y)$: it is convex and compact, and its extreme points are the statistics of pure states in the plane of the two axes (Fig. 1). Since $\mathcal{E}_\theta$ is centered at the origin, moving a point radially outward to

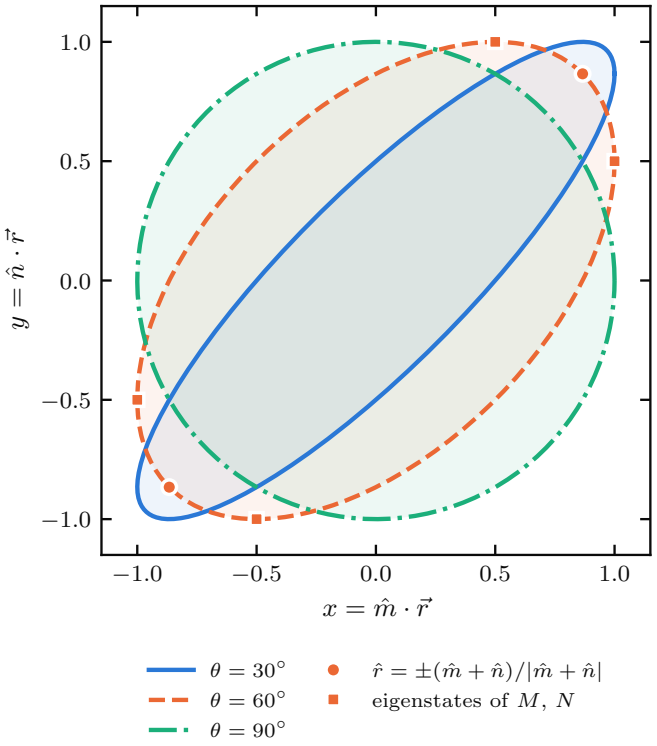


FIG. 1. The ellipse $\mathcal{E}_\theta$ of achievable expectation values $(x, y) = (\langle \hat{m} \cdot \vec{\sigma} \rangle, \langle \hat{n} \cdot \vec{\sigma} \rangle)$ for three Bloch angles. Markers (for $\theta = 60°$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r} \propto \pm(\hat{m} + \hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

$\partial\mathcal{E}_\theta$ multiplies $|x|$ and $|y|$ by the same factor, which makes both p and q more ordered. Every uncertainty measure of the form $\Phi(p) + \Psi(q)$ with Φ and Ψ Schur-concave, in particular every Rényi and Tsallis sum, is therefore minimized on $\partial\mathcal{E}_\theta$, and the optimization problems below are effectively one-dimensional.

IV. PROJECTIVE MEASUREMENTS

A. Direct-sum bound

Theorem 1 ([32, 36]). *Let M, N be projective qubit measurements with maximal overlap c. For every state ρ,*

$$p(\rho) \oplus q(\rho) \;\prec\; \omega_\oplus := \big(1,\ \sqrt{c},\ 1 - \sqrt{c},\ 0\big), \tag{10}$$

and $\omega_\oplus$ is the optimal majorization bound for $\mathcal{P}_\oplus(M, N)$.

The majorization relation (10) is the qubit case of the direct-sum relation $p \oplus q \prec \{1\} \oplus W$ of Rudnicki *et al.* [32]: for $d = 2$ their vector $W = (s_1, s_2 - s_1)$ has $s_1 = \sqrt{c}$ and $s_2 = 1$. Its optimality, and the states that saturate its partial sums, were obtained by Li and Qiao in the majorization lattice [36]. We give a short proof based on Lemma 2.

Proof. We compute $S_k = \max_\rho L_k(p \oplus q)$ for $k = 1, \ldots, 4$. Clearly $S_1 \le 1$ and $S_4 = 2$. Since all entries are nonnegative, $S_3 = 2 - \min_\rho \min_i (p \oplus q)_i \le 2$. Both bounds are attained by an eigenstate of M, for which $p = (1, 0)$. For $k = 2$, the two largest entries either belong to the same distribution, with sum 1, or are of the form $p_i + q_j$. By Lemma 2,

$$p_i + q_j = 1 + \tfrac{1}{2}(\pm x \pm y) \le 1 + \tfrac{1}{2} \max\{|\hat{m} + \hat{n}|, |\hat{m} - \hat{n}|\}. \tag{11}$$

With $\hat{m} \cdot \hat{n} = \cos\theta \ge 0$ the maximum is $|\hat{m} + \hat{n}| = \sqrt{2(1 + \cos\theta)} = 2\cos\frac{\theta}{2} = 2\sqrt{c}$, so

$$S_2 = 1 + \sqrt{c}, \tag{12}$$

attained by the bisector state $\vec{r} = (\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$. This is the qubit case of the Landau–Pollak-type inequality $p_i + q_j \le 1 + \sqrt{c}$ underlying Deutsch's relation [12, 52]. The increments of $(S_1, S_2, S_3, S_4) = (1, 1 + \sqrt{c}, 2, 2)$ give $\omega_\oplus$, which is non-increasing because $1 \ge \sqrt{c} \ge 1 - \sqrt{c} \ge 0$. Lemma 1 completes the proof. □

B. Tensor-product bound

Theorem 2 ([30, 31]). *Under the assumptions of Theorem 1, for every state ρ,*

$$p(\rho) \otimes q(\rho) \;\prec\; \omega_\otimes := \big(w,\ 1 - w,\ 0,\ 0\big), \tag{13}$$

where $w = \frac{1}{4}\big(1 + \sqrt{c}\big)^2$, and $\omega_\otimes$ is the optimal majorization bound for $\mathcal{P}_\otimes(M, N) = \{p(\rho) \otimes q(\rho)\}$.

The vector $\omega_\otimes$ coincides with the vector $Q^{(1)} = (C^2, 1 - C^2, 0, 0)$, $C = (1 + \sqrt{c})/2$, of Puchała *et al.* [31], which for $d = 2$ exhausts their hierarchy of bounds. It is also the vector of Friedland *et al.* [30], whose entries are the increments of the maximal partial sums S_k; for qubits these increments are non-increasing, so that $\omega_\otimes$ is the least upper bound.

Proof. By the arithmetic–geometric mean inequality and Eq. (12), $p_i q_j \le \big(\frac{p_i + q_j}{2}\big)^2 \le \big(\frac{1 + \sqrt{c}}{2}\big)^2 = w$, with equality for the bisector state, for which $p_+ = q_+ = (1 + \sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M, $p \otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \ge 1/\sqrt{2}$ we have $w \ge \frac{1}{2}$, so the increments $(w, 1 - w, 0, 0)$ are non-increasing, and Lemma 1 applies. □

C. Attainability

Neither optimal bound is the statistics of a state unless the measurements commute.

Proposition 1. *(i) There is a state with $(p \oplus q)^\downarrow = \omega_\oplus$ if and only if $c = 1$, i.e., M and N commute. For $c < 1$, the maxima S_1, S_3 are attained only by the eigenstates of M and N, and S_2 only by the bisector states $\vec{r} = \pm(\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$ (for $\theta = \pi/2$ also by $\vec{r} = \pm(\hat{m} - \hat{n})/|\hat{m} - \hat{n}|$). (ii) There is a state with $(p \otimes q)^\downarrow = \omega_\otimes$ if and only if $c = 1$.*

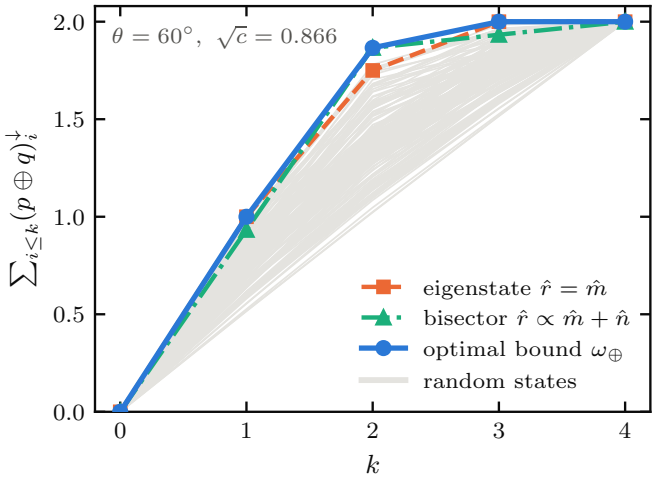


FIG. 2. Lorenz curves of $p \oplus q$ for $\theta = 60°$. Grey: 150 random states. The optimal bound $\omega_\oplus$ (blue) is the smallest concave curve above all of them; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it. A similar plot for $\theta = 90°$ is Fig. 1(a) of Ref. [36].

Proof. (i) $(p \oplus q)^\downarrow_1 = 1$ forces ρ to be an eigenstate of M or N, say $\vec{r} = \pm\hat{m}$. Then $q = \left(\frac{1\pm\cos\theta}{2}, \frac{1\mp\cos\theta}{2}\right)$ and $(p\oplus q)^\downarrow = (1, c, 1-c, 0)$, which equals $\omega_\oplus$ iff $c = \sqrt{c}$, i.e., $c = 1$. The statement about S_2 follows from the equality condition in Cauchy–Schwarz and from $|\hat{m}-\hat{n}| < |\hat{m}+\hat{n}|$ for $\theta < \pi/2$. (ii) If $c = 1$, a common eigenstate of M and N gives $p \otimes q = (1, 0, 0, 0) = \omega_\otimes$. If $c < 1$, the largest entry of $p \otimes q$ equals w only if both inequalities in the proof of Theorem 2 are equalities, i.e., $p_i = q_j$ and $p_i + q_j = 1 + \sqrt{c}$. The state then attains the direct-sum maximum $S_2 = 1 + \sqrt{c}$, so by (i) it is one of the states listed there, and both of its outcome distributions are $(a, 1-a)$ up to order, with $a = (1 + \sqrt{c})/2$. Hence $L_2(p \otimes q) = a^2 + a(1-a) = a < 1 = L_2(\omega_\otimes)$. □

Proposition 1 makes precise the sense in which $\omega_\oplus$ and $\omega_\otimes$ are *descriptions* of the achievable sets rather than the statistics of a particular state: each is a supremum in the majorization lattice, and the partial sums of $\omega_\oplus$ are saturated by two disjoint families of states (Fig. 2). Demanding certainty about M lowers the largest attainable $L_2(p \oplus q)$ from $1 + \sqrt{c}$ to $1 + c$; this deficit, $\sqrt{c}(1 - \sqrt{c})$, vanishes only for commuting measurements.

V. ENTROPIC UNCERTAINTY RELATIONS

Let $h(t) = -t \log_2 t - (1-t)\log_2(1-t)$ be the binary entropy.

A. Bounds for the Rényi and Tsallis families

Corollary 1 ([32]). *For every concave ϕ on $[0, 1]$, $\sum_i \phi(p_i) + \sum_j \phi(q_j) \ge \phi(1) + \phi(\sqrt{c}) + \phi(1 - \sqrt{c}) + \phi(0)$. In particular,*

$$H(p) + H(q) \ \ge\ h(\sqrt{c}), \tag{14}$$

$$T_\gamma(p) + T_\gamma(q) \ \ge\ \frac{1 - c^{\gamma/2} - (1-\sqrt{c})^\gamma}{\gamma - 1} \qquad (\gamma > 0), \tag{15}$$

and, for $0 < \alpha \le 1$,

$$H_\alpha(p) + H_\alpha(q) \ \ge\ H_\alpha(\sqrt{c}, 1 - \sqrt{c}), \tag{16}$$

where $H_\alpha(\sqrt{c}, 1-\sqrt{c}) = \log_2\left[c^{\alpha/2} + (1-\sqrt{c})^\alpha\right]/(1-\alpha)$.

Proof. The first statement is Karamata's inequality applied to Theorem 1. Eqs. (14) and (15) follow with $\phi(t) = -t\log_2 t$ and $\phi_\gamma(t) = (t - t^\gamma)/(\gamma - 1)$, since $\sum_i p_i = \sum_j q_j = 1$ gives $T_\gamma(p) + T_\gamma(q) = \sum_k \phi_\gamma\big((p\oplus q)_k\big)$. Eq. (16) is the qubit case of Theorem 2 of Ref. [32]; we recall the short argument. For $\alpha < 1$ write $\sum_i p_i^\alpha = 1 + X$ and $\sum_j q_j^\alpha = 1 + Y$ with $X, Y \ge 0$. Then

$$\begin{aligned} H_\alpha(p) + H_\alpha(q) &= \frac{\log_2[(1+X)(1+Y)]}{1-\alpha} \\ &\ge \frac{\log_2(1 + X + Y)}{1-\alpha}, \end{aligned}$$

and $1 + X + Y = \sum_k (p \oplus q)_k^\alpha - 1 \ge \sum_k (\omega_\oplus)_k^\alpha - 1 = c^{\alpha/2} + (1-\sqrt{c})^\alpha$ by Schur-concavity of $v \mapsto \sum_k v_k^\alpha$. □

Corollary 2 ([30, 31]). *For every $\alpha \in [0, \infty]$,*

$$H_\alpha(p) + H_\alpha(q) \ \ge\ H_\alpha(\omega_\otimes) = \frac{\log_2\left[w^\alpha + (1-w)^\alpha\right]}{1-\alpha}. \tag{17}$$

For $\alpha = 1$ this reads $H(p) + H(q) \ge h(w)$; for $\alpha = \infty$ it gives $H_\infty(p) + H_\infty(q) \ge -2\log_2 \frac{1+\sqrt{c}}{2}$, which is tight (bisector state); since $H \ge H_\infty$, it contains Deutsch's Shannon bound [12], which has the same right-hand side.

As noted in Sec. II, the two forms suit different families of entropies. Where both apply, the direct sum is stronger, but it cannot be extended to all orders. That the direct-sum Rényi relation does not hold for $\alpha > 1$ was stated in Ref. [32] (Sec. III); part (ii) below gives an explicit counterexample:

Proposition 2. *(i) For $0 < \alpha \le 1$, $H_\alpha(\sqrt{c}, 1-\sqrt{c}) \ge H_\alpha(\omega_\otimes)$, with equality iff $c = 1$; in particular $h(\sqrt{c}) \ge h(w)$. (ii) For $c < 1$ the direct-sum form (16) fails at $\alpha = \infty$: $\min_\rho[H_\infty(p) + H_\infty(q)] = -2\log_2\frac{1+\sqrt{c}}{2} < -\frac{1}{2}\log_2 c = H_\infty(\sqrt{c}, 1-\sqrt{c})$.*

Proof. (i) Since $w - \sqrt{c} = \frac{1}{4}(1-\sqrt{c})^2 \ge 0$, we have $(\sqrt{c}, 1-\sqrt{c}) \prec (w, 1-w)$, with equality iff $c = 1$; apply strict Schur-concavity. The analogous statement in any dimension is proven in Ref. [32]. (ii) With $t = \sqrt{c}$, $\left(\frac{1+t}{2}\right)^2 - t = \left(\frac{1-t}{2}\right)^2 > 0$ for $t < 1$; the minimum is the tight bound of Corollary 2. □

B. When are the bounds tight?

Theorem 3. *Let $c < 1$. (i) For every continuous, strictly Schur-concave Φ,*

$$\min_{\rho} \Phi(p \oplus q) > \Phi(\omega_{\oplus}), \qquad \min_{\rho} \Phi(p \otimes q) > \Phi(\omega_{\otimes}). \quad (18)$$

Consequently the bounds (14), (15) and (16), and the bounds (17) with $0 < \alpha < \infty$, are all strict. (ii) The bounds (17) of order $\alpha = 0$ and $\alpha = \infty$ are tight; they are attained by the eigenstates of M or N and by the bisector states, respectively.

Proof. (i) The achievable statistics form a compact set, so the minimum of Φ is attained at some state ρ^*. By Theorem 1 and Proposition 1(i), $p \oplus q(\rho^*) \prec \omega_{\oplus}$ with $(p \oplus q)^{\downarrow} \neq \omega_{\oplus}$, hence $\Phi(p \oplus q(\rho^*)) > \Phi(\omega_{\oplus})$. The tensor product is analogous, by Theorem 2 and Proposition 1(ii). The Shannon and Tsallis sums are strictly Schur-concave functions of $p \oplus q$, and $H_\alpha(p) + H_\alpha(q) = H_\alpha(p \otimes q)$ with H_α strictly Schur-concave for $0 < \alpha < \infty$. For (16) with $\alpha < 1$, apply the argument to $v \mapsto \sum_k v_k^\alpha$ in the proof of Corollary 1. (ii) An eigenstate of M gives $H_0(p) + H_0(q) = 0 + 1 = H_0(\omega_{\otimes})$, and a bisector state gives $\max_i p_i \max_j q_j = w$. □

The proof of (i) uses only that the achievable statistics form a compact set whose least upper bound is not attained. It therefore applies whenever the optimal bound is not attained: to every pair of qubit POVMs whose effects do not commute (Proposition 3) and to both bounds for binary projective measurements in any dimension with $c < 1$ (Theorem 5).

Theorem 3 makes precise in what sense the majorization bounds are optimal. As majorization bounds they cannot be improved (Theorems 1 and 2); yet, because the supremum is not the statistics of any state, the bound it induces is strict for every strictly Schur-concave measure. For such measures the optimum has to be found measure by measure, which is what entropy-specific relations do; only at the extreme orders $\alpha = 0$ and $\alpha = \infty$ does the majorization bound reach it.

C. Comparison with the Maassen–Uffink relation and with the exact optimum

Figure 3 compares the Shannon bounds with MU and with the exact minimum of $H(p)+H(q)$, which is known for qubits [40, 41]. By Sec. III the minimum lies on the great circle through $\hat{m}$ and $\hat{n}$. For $\theta \le \bar{\theta}$ it is attained at the bisector states and equals $2h\big(\frac{1+\sqrt{c}}{2}\big)$ [41]. At $\bar{\theta}$ the bisector turns from a local minimum into a local maximum along the circle; a second-derivative test shows that this happens when $\sqrt{\bar{c}}\,\operatorname{artanh}\sqrt{\bar{c}} = 1$ [53], i.e., at $\bar{c} \approx 0.6948$ and $\bar{\theta} \approx 67.07°$. For $\bar{\theta} < \theta < \pi/2$ the minimizers bifurcate and the minimum has to be computed numerically [41]. As θ grows they move continuously towards the eigenstates of M and N, which they reach at $\theta = \pi/2$, where MU is tight.

The range in which the bisector is optimal depends on the uncertainty measure, as Zozor *et al.* showed for Rényi entropies [43]. For $c > \frac{1}{2}$ the eigenstates are optimal for $\alpha \le \frac{1}{2}$, the bisector for $\alpha \ge \alpha^\star(c)$, and intermediate states in between, where $\alpha^\star$ decreases from $\alpha^\dagger \approx 1.43$ as $c \to \frac{1}{2}$ to $\frac{1}{2}$ as $c \to 1$; for mutually unbiased measurements the eigenstates remain optimal for all $\alpha < \alpha^\dagger$, and the bisector takes over at $\alpha^\dagger$. In terms of the Bloch angle, the bisector is optimal up to 43° for $\alpha = 3/4$, up to $\bar{\theta} \approx 67°$ for the Shannon entropy and up to 80° for $\alpha = 1.2$, and at all angles for $\alpha \ge \alpha^\dagger$, in line with the tightness of the min-entropy bound. Here $\alpha^\dagger$ solves $2H_\alpha\big(\frac{2+\sqrt{2}}{4}, \frac{2-\sqrt{2}}{4}\big) = 1$, the condition for bisector and eigenstates to perform equally well when the measurements are mutually unbiased. For the max-entropy, for instance, the eigenstates give $\log_2(1 + \sin\theta)$ and the bisector $2\log_2(1 + \sin\frac{\theta}{2})$.

For mutually unbiased measurements ($\theta = 90°$, $c = \frac{1}{2}$) MU is tight (1 bit), while $h(1/\sqrt{2}) \approx 0.872$ and $h(w) \approx 0.844$. As the measurements become more compatible, however, the majorization bounds take over: $h(\sqrt{c}) > -\log_2 c$ for $\sqrt{c} > t^* \approx 0.7585$, i.e., for Bloch angles $\theta < \theta^* \approx 81.3°$ (the tensor bound overtakes MU at $\theta < 79.3°$), as reported numerically in Ref. [34]; earlier majorization bounds [30, 31] were compared with MU and with the optimal qubit bound in Ref. [44]. For the Shannon entropy alone, however, neither majorization bound is the best closed-form relation available. The bound

$$H(p) + H(q) \;\ge\; h\Big(\frac{1 + \sqrt{2c-1}}{2}\Big) \quad (19)$$

of Refs. [40, 44] (see also Ref. [19]), where $2c-1 = \cos\theta$, is stronger than both $h(\sqrt{c})$ and MU at every angle, and it coincides with MU, and hence is tight, at $\theta = 90°$ (Fig. 3); at $\theta = 60°$, for instance, it gives 0.601 bit against 0.568 for $h(\sqrt{c})$. What majorization offers instead is a single bound for all Schur-concave measures at once.

As $c \to 1$ the advantage over MU becomes unbounded, and the direct-sum bound becomes tight in ratio. With $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$ one has $-\log_2 c \approx 2\varepsilon/\ln 2$ and $h(\sqrt{c}) = h(\varepsilon) \approx \varepsilon\log_2(e/\varepsilon)$, whereas the exact minimum is $2h(\varepsilon/2) \approx \varepsilon\log_2(2e/\varepsilon)$. To leading order the ratio of the exact minimum to $h(\sqrt{c})$ is therefore $1 + 1/\log_2(e/\varepsilon)$: it tends to 1, but only logarithmically (1.09 at $\theta = 0.1$ rad, 1.047 at $\theta = 0.003$ rad), while the ratio of $h(\sqrt{c})$ to MU grows as $\frac{1}{2}\ln(8e/\theta^2)$ (3.8 and 7.3). The eigenstates of M, although they saturate S_1 and S_3, give $h(c)$, asymptotically twice the minimum (the ratio approaches 2 slowly; it is 1.67 and 1.82 at the same angles).

Between these limits the strictness of Theorem 3 is far from marginal. Over all angles the direct-sum bound $h(\sqrt{c})$ reaches at least 78% of the exact minimum, the smallest ratio, 0.782, occurring at $\theta \approx 70°$, just beyond $\bar{\theta}$, where the minimizer leaves the bisector; the absolute

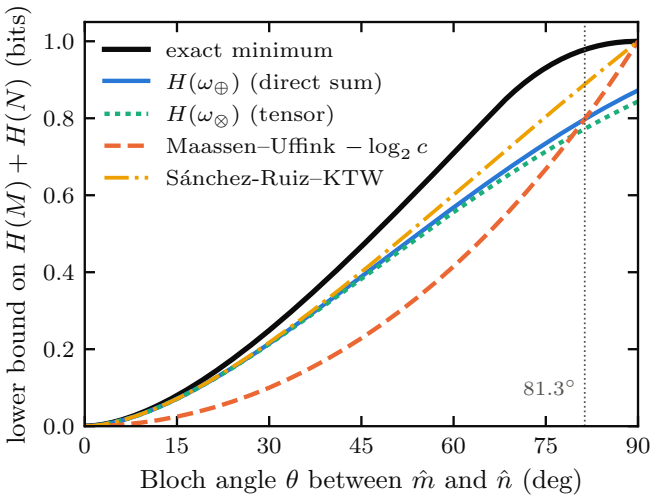


FIG. 3. Lower bounds on $H(M) + H(N)$ as a function of the Bloch angle θ ($c = \cos^2 \frac{\theta}{2}$). Black: exact minimum, equal to $2h\big(\frac{1+\sqrt{c}}{2}\big)$ for $\theta \le \bar{\theta} \approx 67.07°$. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen–Uffink. Yellow (dash-dotted): the closed-form Shannon bound (19) of Refs. [40, 44], which is stronger than both $h(\sqrt{c})$ and MU at every angle. The direct-sum bound is stronger than MU for $\theta < 81.3°$ (dotted vertical line).

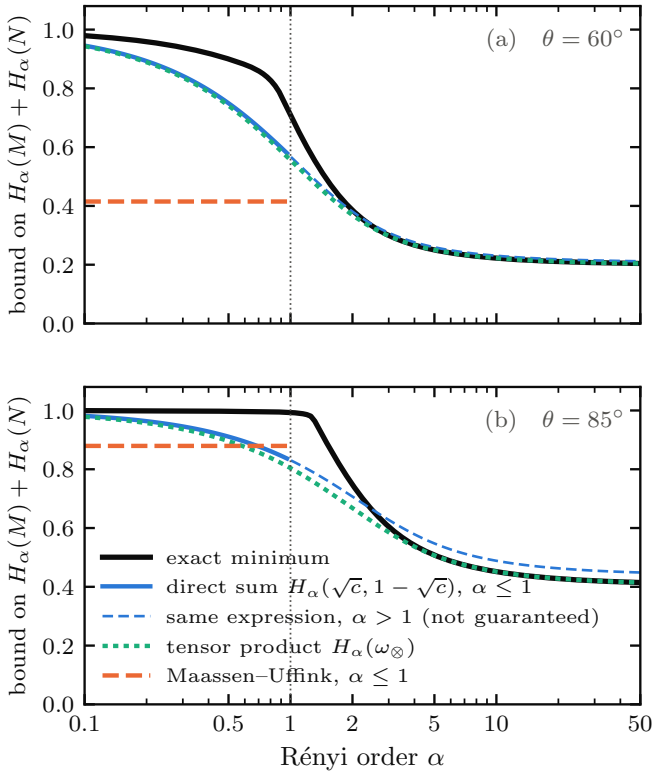


FIG. 4. Bounds on $H_\alpha(M) + H_\alpha(N)$ across the Rényi family for (a) $\theta = 60°$ and (b) $\theta = 85°$. Black: exact minimum. Blue: direct-sum bound $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ of Corollary 1, valid for $\alpha \le 1$; its thin dashed continuation for $\alpha > 1$ is not guaranteed to be a bound and exceeds the exact minimum above $\alpha \approx 2.1$ in (a) and $\alpha \approx 2.5$ in (b). Green (dotted): tensor-product bound $H_\alpha(\omega_\otimes)$ of Corollary 2. Orange (dashed): MU, valid for $\alpha \le 1$. The dotted vertical line marks the Shannon case $\alpha = 1$.

gap is largest, 0.196 bit, at $\theta \approx 74°$, and the tensor-product bound $h(w)$ falls short by up to 0.217 bit. At $\theta = 90°$, where MU is tight, $h(\sqrt{c})$ still misses the exact value 1 bit by 0.128 bit. Wherever a single strictly Schur-concave measure is what matters, such as the Shannon entropy in the analysis of key distribution, this gap is the price of universality, and the entropy-specific optimum should be used.

Figure 4 extends the comparison to the whole Rényi family at $\theta = 60°$ and $\theta = 85°$. For $\alpha \le 1$ MU also applies, because H_α is non-increasing in α. At $\theta = 60°$ both majorization bounds beat MU for all $\alpha \le 1$; at $\theta = 85°$, closer to mutual unbiasedness, they do so only for $\alpha < 0.69$ (direct sum) and $\alpha < 0.58$ (tensor product). For the max-entropy, $\alpha = \frac{1}{2}$, the direct-sum bound reaches 83% and 91% of the exact minimum at the two angles. The expression $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ remains a valid bound somewhat beyond $\alpha = 1$ in these examples, but it fails above $\alpha \approx 2.1$ ($\theta = 60°$) and $\alpha \approx 2.5$ ($\theta = 85°$), in line with Proposition 2(ii). For large α only the tensor-product bound remains, and it converges to the exact minimum as $\alpha \to \infty$. The majorization bounds thus become exact only at the two ends of the family, as Theorem 3 requires.

Finally, MU is one member of a family $H_\alpha(p)+H_\beta(q) \ge -\log_2 c$ with $1/\alpha+1/\beta = 2$ [17], which includes the pair $H_\infty(p) + H_{1/2}(q) \ge -\log_2 c$ of min- and max-entropy. Majorization relations of the form (3) are symmetric under the exchange of p and q and therefore yield such mixed-order relations only indirectly, through the monotonicity of H_α in α. Conditional majorization bounds, which constrain q for a given p, yield them directly: for qubits they give the full uncertainty region of any pair (H_α, H_β) [37].

VI. GENERAL TWO-OUTCOME QUBIT POVMS

For POVMs, the partial sums of the direct-sum relation are largest eigenvalues of sums of effects, $S_k = \max_{|J|=k} \lambda_{\max}\big(\sum_{j\in J} K_j\big)$, where the K_j run over the effects of both measurements [34] (see also Refs. [32, 33, 54]); Baek *et al.* evaluated them for unbiased unsharp qubit observables [34]. For two-outcome qubit POVMs the maximization can be carried out in closed form for arbitrary bias, and the resulting vector turns out to be ordered, hence optimal.

Let $M = \{E_0, E_1\}$ with parameters $(\mu, \eta, \hat{a})$ and $N = \{F_0, F_1\}$ with $(\nu, \zeta, \hat{b})$, as in Eq. (6). Then $p = \frac{1}{2}\big(1+\mu+\eta\,\hat{a}\cdot\vec{r},\ 1-\mu-\eta\,\hat{a}\cdot\vec{r}\big)$, and similarly for q.

Theorem 4. *Define*

$$\kappa = \max\{|\mu| + \eta,\ |\nu| + \zeta\}, \tag{20}$$

$$\Lambda = \max_{s,t=\pm 1} \big[s\mu + t\nu + |s\eta\,\hat{a} + t\zeta\,\hat{b}|\big]. \tag{21}$$

Then $p(\rho) \oplus q(\rho) \prec \omega_{\oplus}(M,N)$ *for all* ρ*, where*

$$\omega_{\oplus}(M,N) = \tfrac{1}{2}\big(1+\kappa,\ 1+\Lambda-\kappa,\ 1+\kappa-\Lambda,\ 1-\kappa\big) \tag{22}$$

is the optimal majorization bound.

Proof. The largest single outcome probability is $\max_\rho \frac{1}{2}\big(1 + |\mu + \eta\,\hat{a}\cdot\vec{r}|\big) = \frac{1}{2}(1 + |\mu| + \eta)$ for M, so $S_1 = \frac{1}{2}(1+\kappa)$; likewise the smallest is $\frac{1}{2}(1-\kappa)$, so $S_3 = 2 - \frac{1}{2}(1-\kappa)$. For S_2, a pair within one distribution sums to 1, while a cross pair gives, by the Cauchy–Schwarz inequality as in Eq. (8),

$$\max_\rho \big[1 + \tfrac{1}{2}\big(s(\mu + \eta\,\hat{a}\cdot\vec{r}) + t(\nu + \zeta\,\hat{b}\cdot\vec{r})\big)\big] = 1 + \tfrac{1}{2}\big[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}|\big]. \tag{23}$$

Choosing $s = \operatorname{sgn}\mu$ and $t = \operatorname{sgn}\nu$ (either sign if zero) shows $\Lambda \geq 0$, so $S_2 = 1 + \frac{1}{2}\Lambda$. The increments of (S_k) are the entries of Eq. (22). They are non-increasing iff $\kappa \leq \Lambda \leq 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda \leq |\mu| + |\nu| + \eta + \zeta \leq 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M; then $S_2 \geq S_1 + \max_j q_j(\rho^*) \geq S_1 + \frac{1}{2}$, i.e., $\Lambda \geq \kappa$. Lemma 1 completes the proof. □

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$ and Eq. (22) reduces to Eq. (10). For unbiased measurements ($\mu = \nu = 0$), $\Lambda = (\eta^2 + \zeta^2 + 2\eta\zeta|\hat{a}\cdot\hat{b}|)^{1/2}$, in agreement with Ref. [34]. If moreover $\eta = \zeta$, then $\omega_{\oplus}(M,N) = \eta\,\omega_{\oplus} + \frac{1-\eta}{2}(1,1,1,1)$, where $\omega_{\oplus}$ is the vector (10) of the sharp measurements along $\hat{a}$ and $\hat{b}$: such measurements are noisy versions of sharp ones [34, 55], and the achievable statistics, their supremum and the saturating states are those of the sharp case mixed with the uniform vector. For unequal sharpness or nonzero bias, $\omega_{\oplus}(M,N)$ is not of this form, and the relevant geometric quantities are the lengths $|\eta\hat{a} \pm \zeta\hat{b}|$ rather than the overlap c.

As for projective measurements, the bound is not attained when the effects do not commute; with bias, it may not be attained even when they do.

Proposition 3. *Let* M *and* N *be as in Theorem 4. (i) If* E_0 *and* F_0 *do not commute, i.e.,* $\eta\zeta\,\hat{a}\times\hat{b} \neq 0$*, no state attains* $\omega_{\oplus}(M,N)$*. (ii) If* $\mu = \nu = 0$ *and* E_0 *and* F_0 *commute, a state attains* $\omega_{\oplus}(M,N)$*. (iii) With bias, commuting effects need not attain it: for* $\hat{b} = \hat{a}$*,* $(\mu,\eta) = (0.3, 0.6)$ *and* $(\nu,\zeta) = (-0.8, 0.15)$*,* S_1 *is attained only at* $\vec{r} = -\hat{a}$ *and* S_2 *only at* $\vec{r} = \hat{a}$*.*

Proof. (i) Suppose a state attains $S_1 = \frac{1}{2}(1+\kappa)$ through an outcome of M, so that $\kappa = |\mu| + \eta$; the case of N is symmetric. Since $\eta > 0$, this forces $\vec{r} = s_0\hat{a}$ with $s_0\mu = |\mu|$. There the two largest entries of $p \oplus q$ sum to $1 + \frac{1}{2}\max_t\big[|\mu| + \eta + t\nu + ts_0\zeta\,\hat{a}\cdot\hat{b}\big]$, whereas the term $s = s_0$ of Λ gives $S_2 \geq 1 + \frac{1}{2}\big[|\mu| + t\nu + |\eta\hat{a} + ts_0\zeta\hat{b}|\big]$ for each t. By Cauchy–Schwarz, $|\eta\hat{a} + ts_0\zeta\hat{b}| > \eta + ts_0\zeta\,\hat{a}\cdot\hat{b}$ unless $\zeta\hat{b} \parallel \hat{a}$, so no state attains both S_1 and S_2. (ii) If one measurement is trivial, say $\eta = 0$, then $\Lambda = \kappa = \zeta$, and $\vec{r} = \hat{b}$ gives $(p \oplus q)^{\downarrow} = \frac{1}{2}(1+\zeta, 1, 1, 1-\zeta) = \omega_{\oplus}(M,N)$. Otherwise, up to relabeling the outcomes of N, $\hat{b} = \hat{a}$, so that $\Lambda = \eta + \zeta$ and $\kappa = \max\{\eta,\zeta\}$, and $\vec{r} = \hat{a}$ gives $(p \oplus q)^{\downarrow} = \frac{1}{2}(1+\kappa, 1+\Lambda-\kappa, 1+\kappa-\Lambda, 1-\kappa)$. (iii) Here $\kappa = 0.95$ is reached only by the second outcome of N at $\vec{r} = -\hat{a}$, and $\Lambda = 1.55$ only by $(s,t) = (1,-1)$ at $\vec{r} = \hat{a}$. □

Bias and unequal sharpness arise naturally in photon counting. If the two outputs of a polarizing beam splitter are monitored by detectors of efficiencies η_H and η_V, and no-click events are assigned to either outcome at random, the effect of outcome H has the form (6) with $\hat{a} = \hat{z}$, $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$. A mismatch of detection efficiencies, a known concern in quantum key distribution [56], thus appears as a bias. A single click/no-click detector behind a polarizer, with no-click events counted as the second outcome, gives $\mu = \eta_d - 1$ and $\eta = \eta_d$, so that $|\mu| + \eta = 1$.

Uncertainty versus incompatibility.—For unbiased POVMs the vectors $\eta\hat{a} \pm \zeta\hat{b}$ that determine Λ also decide joint measurability: the pair is jointly measurable if and only if $|\eta\hat{a} + \zeta\hat{b}| + |\eta\hat{a} - \zeta\hat{b}| \leq 2$ [10, 55]; for biased POVMs the criterion is more involved and no longer depends on these vectors alone [57–59]. The two notions are nevertheless distinct. For orthogonal axes and $\eta = \zeta \leq 1/\sqrt{2}$ the measurements are jointly measurable, yet $S_2 = 1 + \eta/\sqrt{2}$ stays below the value $1+\eta$ that would be reached if both outcome distributions could be sharpened at once, and $\omega_{\oplus}(M,N)$ is not attained (Proposition 3). Attainment requires commuting effects, whether or not the pair is jointly measurable, and with bias even commuting effects need not attain the bound: in the example of Proposition 3(iii) the two readouts of the same axis favor opposite ends of it, a trade-off of classical origin. Majorization uncertainty relations therefore quantify preparation uncertainty, which persists in the absence of measurement incompatibility.

As a corollary, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ with $\omega = \omega_{\oplus}(M,N)$ for all two-outcome qubit POVMs; by Proposition 3 and Theorem 3 this bound is strict whenever E_0 and F_0 do not commute. Figure 5 shows that it tracks the exact minimum as a function of sharpness, becoming exact in the trivial limit $\eta \to 0$, and of bias, where the gap grows with the bias at 45° but not at 90°.

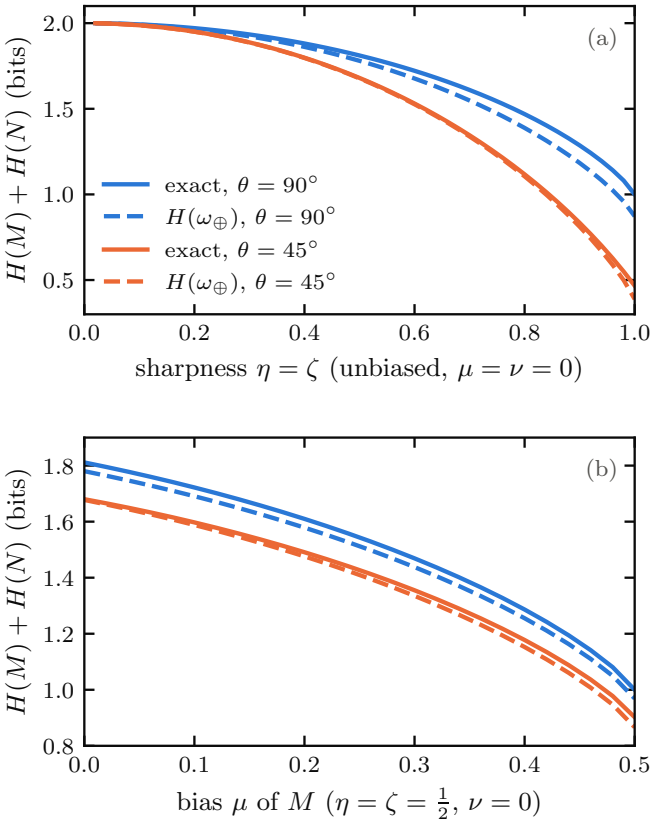


FIG. 5. Exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 4 (dashed) for Bloch angles 90° and 45°. (a) Unbiased measurements of equal sharpness ($\mu = \nu = 0$, $\eta = \zeta$). (b) M with sharpness $\eta = \frac{1}{2}$ and bias μ; N unbiased with $\zeta = \frac{1}{2}$.

VII. BINARY PROJECTIVE MEASUREMENTS IN DIMENSION d

The qubit analysis already contains the general binary projective case.

Theorem 5. *Let $M = \{P, \mathbb{1} - P\}$ and $N = \{Q, \mathbb{1} - Q\}$ be projective measurements on $\mathbb{C}^d$, write $P_1 = P$, $P_2 = \mathbb{1} - P$, $Q_1 = Q$, $Q_2 = \mathbb{1} - Q$, and let $c = \max_{i,j} \|P_i Q_j\|^2$, where $\|\cdot\|$ is the operator norm. Then Theorems 1 and 2 hold verbatim, and $\omega_\oplus$ and $\omega_\otimes$ are attained if and only if $c = 1$.*

Proof. By the eigenvalue formula of Sec. VI, S_k for the direct sum is the largest eigenvalue of a sum of k of the four projectors. Hence $S_1 = 1$, $S_3 = 2$, since any three of them include $P_1 + P_2 = \mathbb{1}$ or $Q_1 + Q_2 = \mathbb{1}$, and $S_2 = \max_{i,j} \lambda_{\max}(P_i + Q_j) = 1 + \sqrt{c}$, since $\lambda_{\max}(P+Q) = 1 + \|PQ\|$ for any two projectors [60]. Moreover $c \geq \frac{1}{2}$: if $Q_1 \neq 0$ and ψ is a unit vector in its range, then $\|P_1\psi\|^2 + \|P_2\psi\|^2 = 1$, so $\max_i \|P_i Q_1\|^2 \geq \frac{1}{2}$ (and if $Q_1 = 0$, then $Q_2 = \mathbb{1}$ and $c = 1$). The increments therefore give $\omega_\oplus$, which is ordered because $1 \geq \sqrt{c} \geq 1 - \sqrt{c}$, so Lemma 1 yields Theorem 1. For the tensor product, $p_i q_j \leq \left(\frac{p_i + q_j}{2}\right)^2 \leq w$, with equality at a top eigenvector of $P_i + Q_j$ with $\|P_i Q_j\|^2 = c$. Indeed, by Jordan's lemma [60], which is also the standard tool for entropic uncertainty relations of binary measurements [44], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces of dimension one or two that are invariant under P and Q; on a two-dimensional subspace with overlap c, the bisector state of the qubit measurements that P and Q induce there gives $p_i = q_j = (1 + \sqrt{c})/2$, and for $c = 1$ a common eigenvector gives $p_i = q_j = 1$. Hence $S_1 = w$, while a state in the range of P gives $S_2 = S_3 = S_4 = 1$; since $c \geq \frac{1}{2}$ implies $w \geq \frac{1}{2}$, the increments $(w, 1-w, 0, 0)$ are ordered, and Theorem 2 follows.

If $c < 1$, a state with $L_1(p \oplus q) = 1$ lies in the range of some P_i (or Q_j); then $q_j = \mathrm{tr}(\rho P_i Q_j P_i) \leq \|P_i Q_j\|^2 \leq c$ for all j, so $L_2(p \oplus q) \leq 1 + c < 1 + \sqrt{c}$ and $\omega_\oplus$ is not attained. Likewise, $L_2(p \otimes q) = 1$ requires two vanishing entries of $p \otimes q$, hence such a state, at which $L_1(p \otimes q) \leq c < w$, so $\omega_\otimes$ is not attained either. If $c = 1$, the ranges of some P_i and Q_j intersect, and a common eigenvector attains $\omega_\oplus = (1, 1, 0, 0)$ and $\omega_\otimes = (1, 0, 0, 0)$. □

In particular, all bounds of Sec. V hold for binary projective measurements in any dimension with $c = \max_{i,j} \|P_i Q_j\|^2$, the min-entropy bound remains tight, and for $c < 1$ the bounds for strictly Schur-concave measures are strict (Theorem 3). Equivalently, $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$ with $A = 2P - \mathbb{1}$ and $B = 2Q - \mathbb{1}$. Indeed, on a two-dimensional block, where P and Q act as qubit projectors with axes $\hat{m}_b$ and $\hat{n}_b$, $\{A, B\} = 2(\hat{m}_b \cdot \hat{n}_b)\mathbb{1}$ and $c_b = \frac{1}{2}(1 + |\hat{m}_b \cdot \hat{n}_b|)$, while $\{A, B\} = \pm 2$ on a one-dimensional block. For qubits this is the relation between the overlap and the effective anticommutator of Ref. [44]; it is closely related to the anticommutator norm formula $\|PQ + QP\| = \|PQ\| + \|PQ\|^2$ [61].

The argument for the direct sum uses only the eigenvalue formula, so it carries over to binary POVMs in any dimension.

Corollary 3. *Let $M = \{E_1, E_2\}$ and $N = \{F_1, F_2\}$ be two-outcome POVMs on $\mathbb{C}^d$, let S_1 be the largest eigenvalue of the four effects, and let $S_2 = \max\{1, \max_{i,j} \lambda_{\max}(E_i + F_j)\}$. Then*

$$p(\rho) \oplus q(\rho) \;\prec\; \left(S_1,\; S_2 - S_1,\; 1 + S_1 - S_2,\; 1 - S_1\right) \qquad \forall \rho, \tag{24}$$

and this vector is the optimal majorization bound.

Proof. By the eigenvalue formula, S_1 and S_2 are the maximal partial sums, and $S_3 = 1 + S_1$, since any three effects include $E_1 + E_2 = \mathbb{1}$ or $F_1 + F_2 = \mathbb{1}$. The increments are non-increasing iff $S_1 + \frac{1}{2} \leq S_2 \leq 2S_1$. The upper inequality holds because $\lambda_{\max}$ is subadditive and $S_1 \geq \frac{1}{2}$. For the lower one, a top eigenvector of the effect that gives S_1 is a state at which the other measurement has an outcome of probability at least $\frac{1}{2}$. Lemma 1 completes the proof. □

Theorem 4 is the qubit case, with $S_1 = \frac{1}{2}(1 + \kappa)$ and $S_2 = 1 + \frac{1}{2}\Lambda$, and Theorem 5 the projective case, with

$S_1 = 1$ and $S_2 = 1+\sqrt{c}$. For arbitrary POVMs in any dimension, the eigenvalue formula together with Lemma 1 gives an exact numerical procedure, which we use to verify Theorems 4 and 5 and Corollary 3 independently (Appendix A).

VIII. DISCUSSION

For binary measurements the majorization approach to uncertainty can be carried out in closed form. For a qubit the achievable statistics form an affine image of an ellipse, and a single four-component vector, determined by the overlap c, is their least upper bound in the majorization lattice, for the direct sum as well as for the tensor product. For POVMs the direct-sum bound is determined by κ and Λ, and in any dimension by the largest eigenvalues of the effects and of their pairwise sums (Corollary 3). This is the unified perspective that majorization offers: one vector controls every Schur-concave uncertainty measure at once.

The comparison with entropy-specific relations shows what this unification costs. Because the supremum is not the statistics of any state when the measurements do not commute, the induced bound is strict for every strictly Schur-concave measure, and it becomes exact only at the extreme orders $\alpha = 0$ and $\alpha = \infty$. Against a generic scalar relation such as MU, on the other hand, the majorization bounds are strong. For two projective qubit measurements, both Shannon bounds depend only on the Bloch angle θ between the two measurement axes: the direct-sum bound is $h(\sqrt{c})$ and MU gives $-\log_2 c$, with $c = \cos^2\frac{\theta}{2}$. The two curves cross exactly once, at $\theta^* \approx 81.3°$ ($c \approx 0.575$), the root of $h(\cos\frac{\theta}{2}) = -2\log_2\cos\frac{\theta}{2}$ (Sec. V and Fig. 3). Nothing singles out this angle geometrically; it is simply where the two bounds exchange roles. For $\theta < \theta^*$, that is, unless the measurements are close to mutually unbiased, the direct-sum Shannon bound is the stronger of the two, and it becomes asymptotically tight in ratio as the measurements approach compatibility, precisely where MU is weakest. Only in the narrow window $\theta^* < \theta \leq 90°$ next to mutually unbiased measurements ($\theta = 90°$), for which MU is tight, does MU give the better bound; for the tensor-product bound the crossover lies at 79.3°. For the Shannon entropy alone, the closed-form bound (19) is stronger than both at every angle, and the exact minimum exceeds $h(\sqrt{c})$ by up to 22%; the strength of the majorization bounds lies in their universality. Of the two forms, the direct sum is preferable for $\alpha \leq 1$, while only the tensor product survives for large α.

Relation to quantum optics.—These results contrast instructively with the quadratures of a light mode. There, coherent and squeezed states are minimum-uncertainty states: for their principal quadratures they saturate both the Heisenberg relation and the entropic relation of Białynicki-Birula and Mycielski [15]; moreover, all pure Gaussian states are equivalent under continuous majorization in phase space and are conjectured to majorize every Wigner-positive state [62]. For non-commuting binary measurements no such state exists (Propositions 1 and 3), and the optimal state depends on the uncertainty measure, in line with the finding of Korzekwa and Lostaglio [63] that no qubit state minimizes all information-theoretic uncertainty measures at once. The boundary of $\mathcal{E}_\theta$, traced by the pure states in the plane of the two axes, plays the role of the family of squeezed states, trading predictability of N for that of M. The balanced bisector state, the analogue of a coherent state, minimizes the Shannon sum up to $\bar{\theta} \approx 67°$; for larger angles the minimizer moves towards an eigenstate of either measurement, the analogue of an infinitely squeezed state, and the range in which the balanced state is optimal shrinks for smaller Rényi orders and grows for larger ones (Sec. V and Ref. [43]). The ellipse itself resembles the covariance ellipse of a squeezed vacuum only loosely: it describes the pair of measurements rather than a single state, and although it is a disk for mutually unbiased measurements and flattens as they become compatible, its area $\pi\sin\theta$ shrinks instead of being conserved. A single qubit cannot be squeezed in the usual sense [64], as all its pure states are spin coherent states; the essential difference is algebraic. For quadratures the commutator is a c-number, which makes the Heisenberg bound state independent. For binary observables it is the anticommutator that is, blockwise, a c-number [Eq. (9)], and it fixes $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$ (Sec. VII). The Robertson bound $\Delta A\,\Delta B \geq |\langle(\hat{m}\times\hat{n})\cdot\vec{\sigma}\rangle|$, by contrast, vanishes for every state whose Bloch vector lies in the plane of the two axes, which contains all optimal states. Commutator-based relations are therefore uninformative here, and state-independent descriptions are needed: tight variance relations [47] or, for all Schur-concave measures at once, majorization. The information missing from the Robertson bound is carried by the anticommutator term of the Robertson–Schrödinger relation, $\det\Gamma \geq \langle(\hat{m}\times\hat{n})\cdot\vec{\sigma}\rangle^2$ with Γ from Eq. (9), which every pure state saturates; in the plane of the axes it reduces to $\det\Gamma = 0$, the boundary of $\mathcal{E}_\theta$.

The results apply directly to photonic qubits. Restricted to single photons, the Stokes operators of polarization are the Pauli matrices, and two linear polarization analyzers at a relative angle ϑ correspond to $\theta = 2\vartheta$, so that $c = \cos^2\vartheta$ is Malus's law; the crossover with MU and the angle $\bar{\theta}$ correspond to $\vartheta \approx 40.7°$ and $\vartheta \approx 33.5°$, respectively. The optimal direct-sum bound has been tested in this setting with coherent light, the polarization state being reconstructed from measured Stokes parameters [39]. For many photons the Stokes operators form a collective spin that can be squeezed (polarization squeezing [65, 66]); for a single photon, which cannot be squeezed, only the trade-off along the boundary of $\mathcal{E}_\theta$ remains. Imperfect detectors lead to the unsharp and biased measurements of Theorem 4, with the bias set by the mismatch of detection efficiencies (Sec. VI); by Proposition 3(iii), such biases can create a trade-off even between

two measurements in the same polarization basis.

Several questions remain open. (i) *Tensor bounds for POVMs.* The optimal $\omega_\otimes$ requires maximizing sums of products of affine functions over $\mathcal{E}_\theta$, and we have not found it for any noisy measurements. For unbiased measurements of equal sharpness the AM–GM step of Theorem 2 still gives S_1, but S_3 is no longer attained at the states that saturate the projective bounds: numerically, for orthogonal axes and $\eta = \zeta = 0.8$ it is attained neither by an eigenstate nor by a bisector state, and we have not derived it in closed form. For biased or unequally sharp measurements the AM–GM step itself is no longer tight, and the least concave majorant in Lemma 1 may become non-trivial. (ii) *Binary POVMs in dimension d.* Corollary 3 gives the optimal direct-sum bound in terms of $\lambda_{\max}(E_i + F_j)$. A closed form for this eigenvalue, analogous to $1 + \|P_i Q_j\|$ for projectors, the conditions under which the bound is attained, and the tensor-product bound remain open. (iii) *More measurements.* For qubit bases the eigenvalue formula gives $S_k = \frac{k}{2} + \frac{1}{2}\max_J \left|\sum_{j\in J} \vec{n}_j\right|$, where J runs over sets of k outcomes and $\vec{n}_j$ is the Bloch vector of the projector of outcome j, and Li and Qiao obtained the optimal bound for σ_x, σ_y and σ_z [36]. For random triples of bases we found the increments of S_k always ordered, so that no flattening was needed; whether this holds in general, and what replaces it for POVMs and for the tensor product, is open. (iv) *Mixed orders.* For qubits, conditional majorization bounds yield optimal relations for any pair of orders [37]. Whether state-independent relations of the form (3) can reproduce mixed-order relations such as $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$ remains open. (v) *Binary measurements on light modes.* Which experimentally accessible binary measurements on a light mode have $c < 1$, and what do the bounds of Sec. VII imply for them? In infinite dimension $c = \sup_{i,j} \|P_i Q_j\|^2$ can equal 1 without a common eigenvector. For photon-number parity and displaced parity, whose expectation values sample the Wigner function [67], increasingly squeezed vacua, which contain only even photon numbers, approach simultaneous eigenstates of both, so no non-trivial state-independent relation exists. Tests for coherent states, $\{|\nu\rangle\langle\nu|, \mathbb{1} - |\nu\rangle\langle\nu|\}$, realized by a displacement followed by on/off detection with no-click probability $\pi Q(\nu) = \langle\nu|\rho|\nu\rangle$, with Q the Husimi function, also have $c = 1$, because all states orthogonal to both coherent states are common eigenvectors. Individual partial sums remain informative, however: $\pi[Q(\nu_1) + Q(\nu_2)] \leq 1 + e^{-|\nu_1 - \nu_2|^2/2}$, with equality for the superposition $|\nu_1\rangle + e^{-i\varphi}|\nu_2\rangle$ with $\varphi = \arg\langle\nu_1|\nu_2\rangle$, an optical cat state [68, 69] (the even cat state for $\nu_2 = -\nu_1$), which thus plays the role of the bisector state. Non-trivial relations require $\|\{A, B\}\| < 2$, as for the pseudospin operators of Ref. [70], of which the squeezed vacuum is an eigenstate, or for polarization measurements on single photons.

ACKNOWLEDGMENTS

R. Z. and M. F. C. acknowledge support by the Quantum Science and Technology-National Science and Technology Major Project (Grant No. 2025ZD0301000), the National Key Research and Development Program of China (Grant No. 2023YFA1407100), the Guangdong Province Science, Technology Major Project (Future functional materials under extreme conditions – 2021B0301030005) and the National Natural Science Foundation of China (Grant No. 12574092).

Appendix A: Numerical methods

All analytical results were also checked numerically. We evaluated Lorenz curves, the lattice supremum via least concave majorants [50], the analytic bounds of Theorems 1, 2 and 4 and Corollaries 1–2, the eigenvalue characterization $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j\in J} K_j)$, and exact minima of Rényi and Tsallis entropy sums.

The closed forms were compared with (a) lattice suprema over 1.8×10^5 grid states, for 40 random pairs of qubit bases (Theorems 1 and 2) and 60 random pairs of qubit POVMs (Theorem 4), with maximal deviation $< 10^{-4}$, and (b) the eigenvalue characterization for 200 random qubit POVM pairs and for 150 random pairs of projectors of rank $d/2$ in dimensions $d = 2, 4, 6$, all with $c < 1$ (deviation $< 10^{-14}$). Pairs with a common eigenvector, which always occur for odd d or unequal ranks, have $c = 1$ and were checked separately; the anticommutator form of c was checked for 300 random pairs of projectors of arbitrary rank in dimensions $d \leq 8$, the tensor-product part of Theorem 5 for 50 random pairs in $d = 4, 6$, and Corollary 3 for 300 random pairs of binary POVMs in $d = 2, \ldots, 6$. We also checked Propositions 1–3, Corollaries 1–2, Theorem 3, the critical angle $\bar{\theta}$, the motion of the Shannon minimizer beyond $\bar{\theta}$, the range of Rényi orders in which the bisector or the eigenstates are optimal, the other values quoted in Sec. V, the covariance form of Lemma 2, the jointly measurable example, the detector model and the behavior of the gap in Fig. 5(b) of Sec. VI, the Robertson–Schrödinger statement of Sec. VIII, and the numerical statements in open questions (i) and (iii), the latter for 300 random triples of qubit bases. The statements about light modes in Sec. VIII were checked in a Fock space of dimension 600.

The exact entropy minima are obtained by a dense grid over the great circle in $\mathrm{span}\{\hat{m}, \hat{n}\}$ refined by Nelder–Mead. For Rényi and Tsallis sums this reduction is justified by the radial argument of Sec. III, which applies equally to unbiased unsharp measurements. For biased measurements we minimize only the Shannon sum, which is concave in (x, y) and is therefore minimized on the boundary of $\mathcal{E}_\theta$ as well. To check that the eigenstates are optimal for $\alpha \leq \frac{1}{2}$, where the Rényi sums have a cusp at the eigenstates, we parametrize the circle by an angle φ and write the outcome probabilities as $\cos^2(\varphi/2)$ and

$\sin^2(\varphi/2)$, which avoids the loss of precision in $1-\cos\varphi$.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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